

#ASLI ENGINEERING

Geohash

SWIPE

BY

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Geospatial Internals: Geohash

Redis supports Geospatial Queries

eg: find people near me, within 2km distance



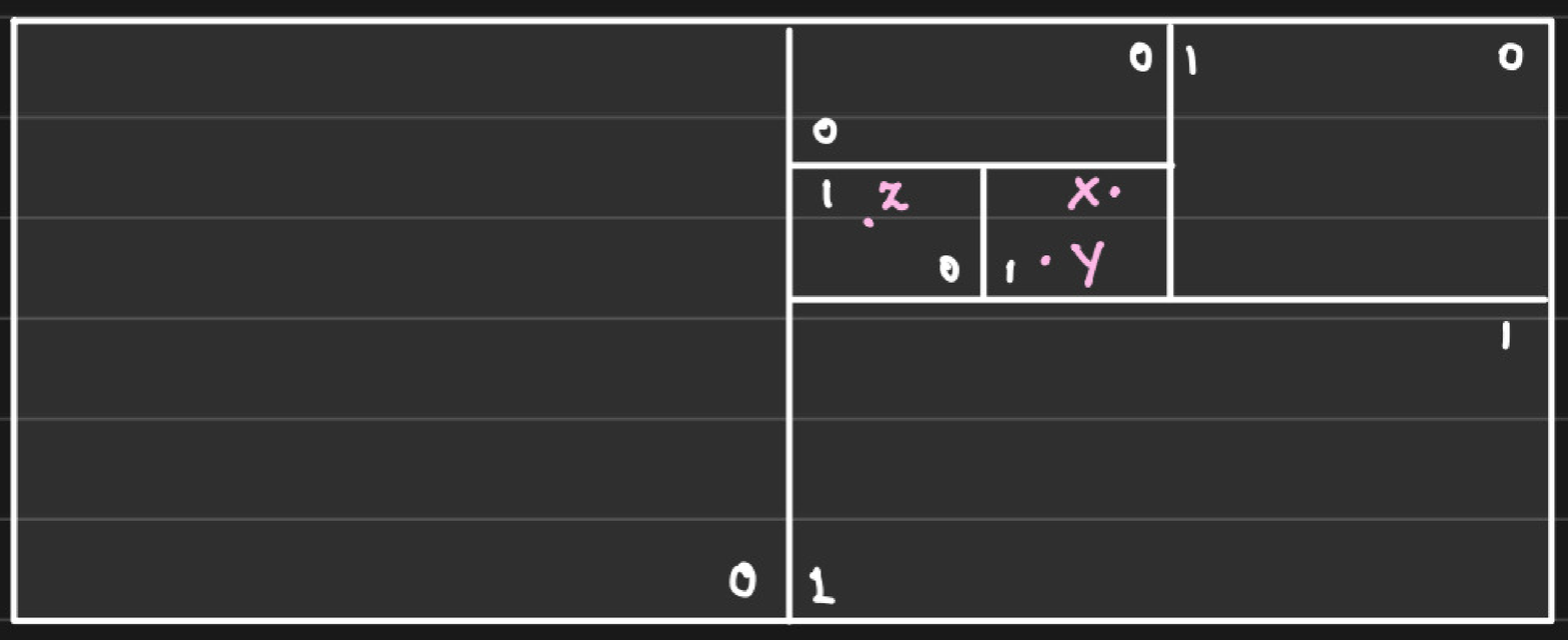
Requirements: The proximity distance is given on the query time

ie. zoom-in and zoom-out capabilities

Redis powers this using Geohash

Geohash: It is a 64 bit integer value that uniquely represents a point in the world

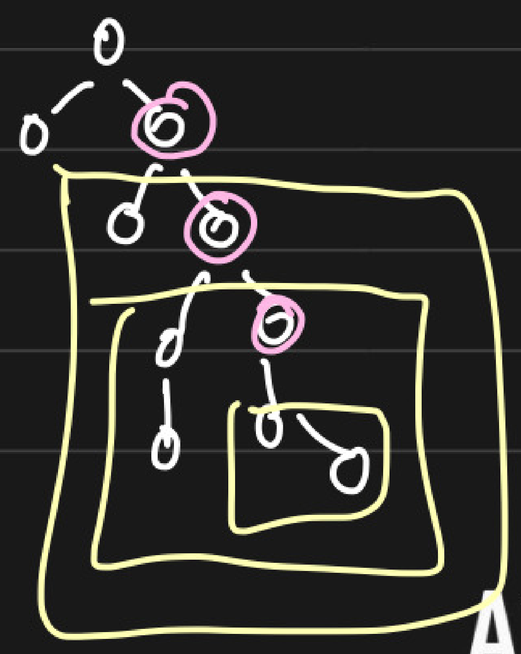
[lat, long] ⟶ f ⟶ g,



x = 10011 1

y = 10011

z = 10010



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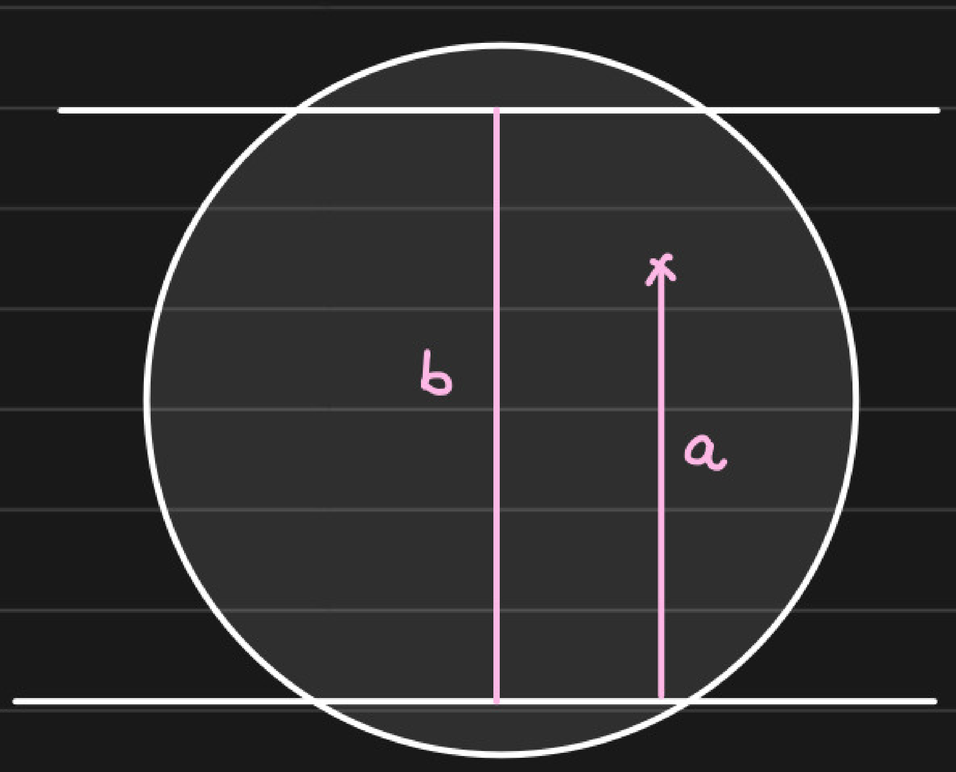
Two entities are closer to each other
if they share a longer geohash prefix

Thus our problem statement has reduced to prefix matching !!

How do we compute ?

We do not need to run a for loop.
instead we can simply take relative positioning

lat offset = a/b



and similarly we do it for longitude

These relative positioning are 32 bit long

Hence, we interleave them bit-by-bit to keep it zoomable

odd bits → longitude
even bits → latitude
} in final 64 bit presentation

Key highlight

- removing bits from right is like zooming out
- making it more precise is zooming in
- modern arch has optimized 64 bit computations

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